

Data Reduction

Live Q&A – 5 March 2025

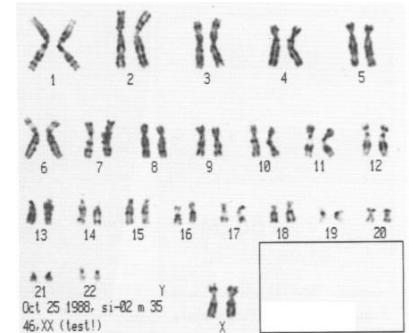
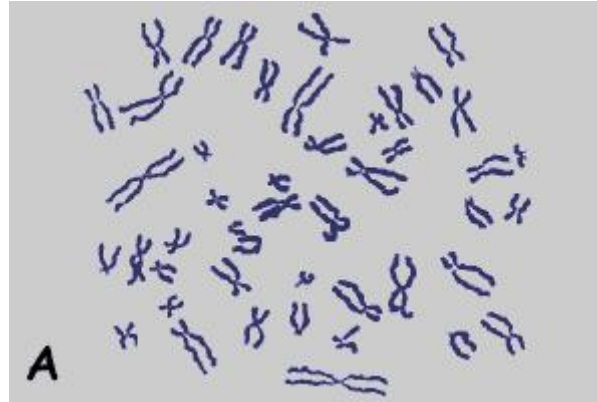
Niall Adams

Overview

- Coursework peek ahead
- Data reduction, various
- From Stochastic Neighbour Embedding to t -SNE
- Q&A

Coursework

- Same deliverables as before
- Hand out: 12 March, 9am
- Hand in: 1 April, 9am



In a study of size and shape relationships, Mosimann [10] measured carapace length, width, and height. Their data, reproduced in Exercise 6.13, Table 6.5, suggests an analysis in terms of logarithms. (Jolicoeur [9] generally suggests a logarithmic transformation in studies of size-and-shape relationships.) Perform a principal component analysis.

The natural logarithms of the dimensions of 24 male turtles have sample mean vector $\bar{x}' = [4.725, 4.478, 3.703]$ and covariance matrix

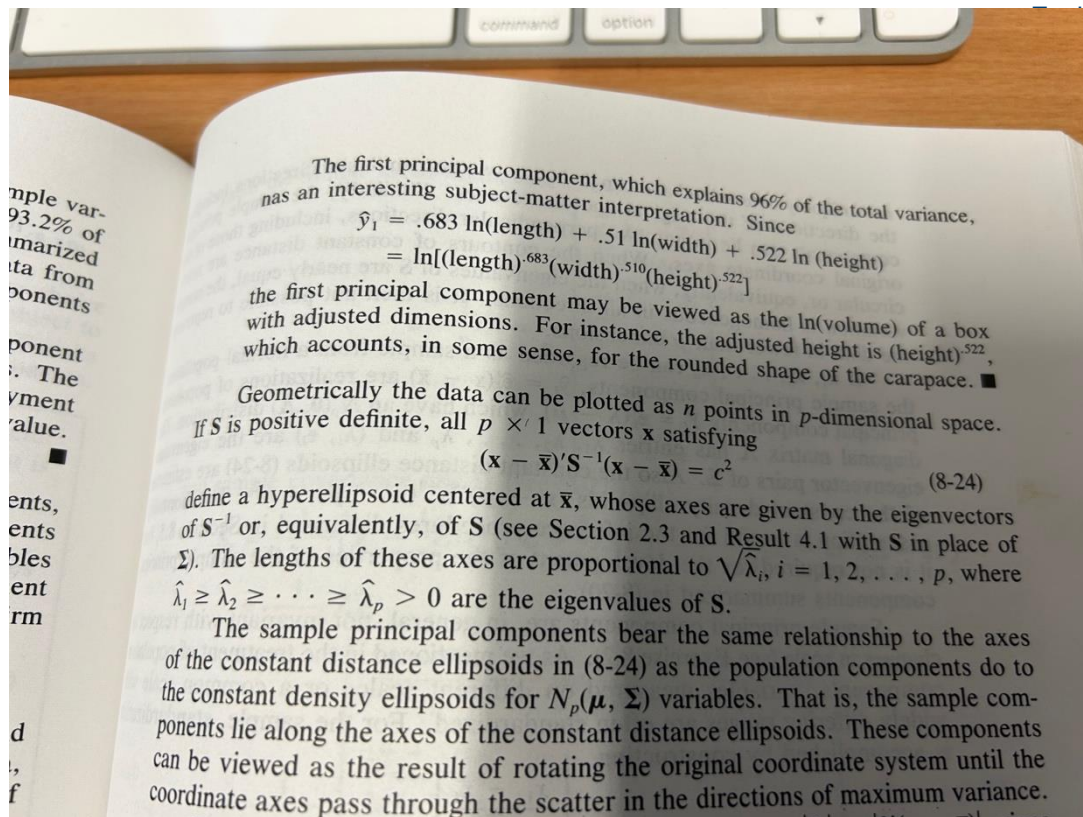
$$S = 10^{-3} \begin{bmatrix} 11.555 & 8.367 & 8.508 \\ 8.367 & 6.697 & 6.264 \\ 8.508 & 6.264 & 7.061 \end{bmatrix}$$

A principal-component analysis yields the following summary.

COEFFICIENTS FOR PRINCIPAL COMPONENTS
(Correlation Coefficients in Parentheses)

Variable	$\hat{e}_1(r_{\hat{e}_1, x_k})$	$\hat{e}_2$	$\hat{e}_3$
ln (length)	.683 (.99)		
ln (width)	.510 (.97)	.162	.712
ln (height)	.522 (.97)	.591	-.624
		-.790	-.321
Variance ($\hat{\lambda}_j$):	24.31×10^{-3}	$.63 \times 10^{-3}$	$.38 \times 10^{-3}$
Cumulative percentage of total variance:	96.0	98.5	100

PCA



Curse of dimensionality

We do not have good intuition about higher dimensions. For example, consider a sphere of radius 1 in D dimensions. Now, suppose we are interested in **the fraction of the volume of the sphere that lies between radius $r = 1 - \varepsilon$ and $r = 1$.**

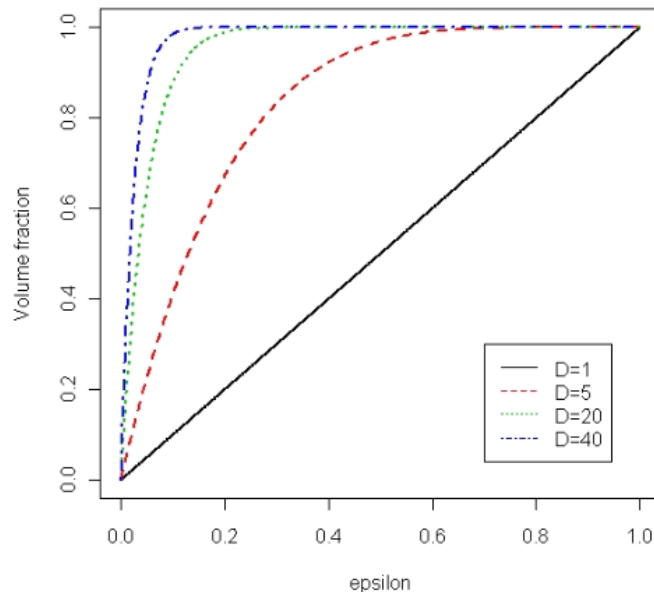
The volume of a sphere in D dimensions must scale with r^D . This volume can be written as

$$V_D(r) = K_D r^D$$

where the constant K_D depends only on D . The required fraction is then

$$\frac{V_D(1) - V_D(1 - \varepsilon)}{V_D(1)} = 1 - \frac{K_D(1 - \varepsilon)^D}{K_D 1^D} = 1 - (1 - \varepsilon)^D$$

Curse of dimensionality



For large D , this tends to 1 even for small values of ε . So, in high dimensional spaces, most of the volume of a sphere is concentrated in a shell near the surface.

Common questions

- Are nonlinear reduction methods always better than linear methods (e.g. PCA)?
 - Nonlinear methods can usually capture linear dependences
 - But often more expensive
 - Also, PCA can generalise easily, e.g. reducing to 3 or 4 dimensions
- Can PCA / MDS / t-SNE be used for feature engineering?

t-SNE additional points

- How do we set the perplexity parameter?
 - t-SNE is claimed to be robust to this choice, but visualisations can vary
 - Generally, lower perplexity emphasises local dissimilarities
- Is the perplexity the only hyperparameter of interest?
- t-SNE gives a different embedding each time you run it
- How do we know if the embedding is a good one?
 - Sometimes by eye is good enough

Any Questions?
